

Evidence-Gated Causal World Models for Continual General Intelligence

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Abstract

World models, large sequence models, modular networks, and neuromorphic hardware address different parts of intelligence. None by itself specifies when a continual agent should acquire information, update a belief, refine a parameter, repair a representation, consolidate a memory, compose known mechanisms, or create persistent structure. We propose an evidence-gated causal world-model architecture that separates high-capacity prediction from authority over long-lived change. Its state combines consequence-indexed predictive state, beliefs over hidden causes and future modes, complementary episodic and semantic memory, reusable mechanisms and skills, an audited query contract, evidence wealth, and measured hardware implementations. Five loops operate at different timescales: reactive control, predictive simulation, active experiment design, recurrent latent reasoning, and structural consolidation. A candidate mechanism must be tested against a declared closure of reuse, composition, refinement, recalibration, branching, representation repair, and corruption; rejection of an incomplete closure remains unresolved. Transformers, joint-embedding objectives, and neuromorphic substrates occupy sequence-operator, learning-objective, and physical-implementation axes and can be combined. We state ten falsification experiments spanning learned nonlinear worlds, memory, reasoning, transfer, and hardware. The architecture is a research hypothesis, not evidence that general intelligence has been achieved. **Keywords:**

world models, causal representation, continual learning, active inference, memory, latent reasoning, neuromorphic computing

1 Why prediction is not enough

Dense world models can master diverse control tasks with one configuration [8], and joint-embedding video models can learn from large action-free corpora before action-conditioned planning [1]. Modular world models seek reusable mechanisms [10, 11], while continual modular systems reuse and expand components [14, 15]. These are major pieces of a general learning system.

An open-ended agent nevertheless faces a distinct control problem. Prediction error can arise because an old mechanism returned, two mechanisms composed, a parameter drifted, uncertainty changed, a sensor corrupted, the state aliased two causes, or a genuinely new mechanism

appeared. A learner that maps all surprise to parameter updates or capacity growth cannot distinguish these possibilities.

We therefore separate two coupled systems:

1. a high-capacity predictor that learns representations, dynamics, language, skills, and plans; and
2. an epistemic governor that controls what may be treated as known, equivalent, safe to forget, or safe to change persistently.

The proposed research question is whether this separation makes continual learning more compact, transferable, and causally correct than prediction-only adaptation.

2 System state

At time t , the agent state is

$$\Omega_t = (s_t, b_t, \mathcal{E}_t, \mathcal{M}_t, \mathcal{K}_t, \mathcal{P}_t, \mathcal{Q}_t, W_t, \mathcal{H}_t), \quad (1)$$

where s_t is learned predictive state; b_t is a calibrated belief over hidden causes and future modes; $\mathcal{E}_t$ is episodic memory; $\mathcal{M}_t$ is a mechanism library; $\mathcal{K}_t$ is consolidated semantic memory; $\mathcal{P}_t$ is a skill library; $\mathcal{Q}_t$ is the audited consequence-query contract; W_t is evidence wealth and authority state; and $\mathcal{H}_t$ is a measured catalogue of physical operator implementations.

Equation 1 is not intended as one monolithic vector. It is a typed interface. Prediction content is separate from the authority to change persistent state.

2.1 Consequence-indexed predictive state

A query $q = (\pi, g, T, c)$ contains an intervention or policy π , bounded consequence functional g , horizon T , and resource envelope c . For history h ,

$$Y_q(h) = g(O_{t+1:t+T}, A_{t:t+T-1}) \mid \text{do}(\pi), H_t = h.$$

Histories are approximately equivalent only when their affordable declared consequences agree:

$$h \sim_{\mathcal{Q}, \varepsilon} h' \iff \sup_{q \in \mathcal{Q}_{\text{aff}}} D_q\{P(Y_q \mid h), P(Y_q \mid h')\} \leq \varepsilon_q. \quad (2)$$

This extends the predictive-state principle [12, 17] with intervention, consequence, and resource indexing. It does not claim that learning the equivalence classes is solved. When new skills enlarge $\mathcal{Q}$, an earlier state may legitimately split.

2.2 Factorized uncertainty

The agent represents

$$U_t = (U^{\text{aleatoric}}, U^{\text{epistemic}}, U^{\text{representation}}, U^{\text{ontology}}, U^{\text{resource}}).$$

Aleatoric uncertainty is residual branching after useful probes. Epistemic uncertainty can shrink through data. Representation uncertainty asks whether the state merged distinguishable histories. Ontology uncertainty asks whether the declared explanation families cover the phenomenon. Resource uncertainty concerns accuracy and cost of a physical implementation. These components trigger different actions; predictive entropy alone cannot do so.

3 Five coupled loops

Reactive control. Fast policies operate from current state and retrieved memory. Adaptation in context is reversible and expires unless consolidated.

Predictive simulation. A dense action-conditioned model predicts distributions or sets of future latent consequences. Simultaneous causes compose through typed operators; mutually exclusive futures retain a separate mode axis. Composition and uncertainty over worlds are not the same operation.

Active experiment design. The agent chooses information actions by

$$q_t^* = \arg \max_q [\mathbb{E} \Delta V(q) + \beta I_q(B; Y) - \lambda^\top c(q) - \rho(q)], \quad (3)$$

where ΔV is expected decision improvement, I_q is information gain about a relevant hidden cause, c is measured cost, and ρ is intervention risk. Information gain is useful only when it can change a decision, improve a model, or repair a representation.

Recurrent latent reasoning. Reasoning applies a shared transition repeatedly to a typed workspace:

$$r^{(k+1)} = F_\theta(r^{(k)}, s_t, \mathcal{R}_t),$$

with retrieved evidence $\mathcal{R}_t$. A halting policy stops when another iteration’s expected value falls below cost or an invariant is met. Recurrent depth can scale latent test-time computation [6], but neural algorithmic reasoning remains fragile under distribution shift and can benefit from causal regularization [2]. The design is therefore a comparative hypothesis, not an assumption of superiority.

Consolidation and structural governance. Episodes propose semantic facts, skills, state splits, refined mechanisms, or new modules. Discovery, confirmation, quarantine, release, rollback, and expiry are distinct. The object earning evidence is bound to the object quarantined.

4 Structural closure and authority

Before mechanism creation, the agent constructs a declared closure

$$\mathfrak{C}_t = \mathfrak{C}_{\text{reuse}} \cup \mathfrak{C}_{\text{compose}} \cup \mathfrak{C}_{\text{refine}} \cup \mathfrak{C}_{\text{recalibrate}} \cup \mathfrak{C}_{\text{branch}} \cup \mathfrak{C}_{\text{repair}} \cup \mathfrak{C}_{\text{corruption}}. \quad (4)$$

One predictable candidate is evaluated against the best explanation in this union. A profile-likelihood instance can use

$$E_t = \frac{A_t}{\max_{c \in \mathfrak{C}_t} \sup_{\theta \in \Theta_c} L_{c, \theta, t}},$$

where A_t is the candidate’s predictable path likelihood. If a declared family upper-bounds the true path likelihood, the statistic is dominated by an oracle likelihood-ratio martingale and inherits an anytime crossing bound from Ville’s inequality [20]. This validity is relative to the declared closure; an omitted cause invalidates a semantic novelty claim.

The cause proposer should therefore return a set rather than one label. A class-conditional conformal set can bound marginal within-class exclusion under exchangeability [4], but it does not certify an open world or an adaptively selected lifetime. Distribution-free out-of-distribution detection also has explicit learnability limits [5]. Missing universe or support coverage maps to abstention.

Authority is typed. Reversible sensing may use a posterior utility decision. Persistent changes consume a lifetime grant and fresh structural, semantic, resource, support, and identity evidence. Evidence earned for a probe cannot be passed to a mechanism-admission interface.

4.1 When a council cannot repair shared error

Council diversity can reduce variance without removing a shared systematic error. For scalar mean expert errors $\mu_1, \dots, \mu_E$, every convex coalition has mean error in

$$\text{conv}\{\mu_e\} = [\min_e \mu_e, \max_e \mu_e].$$

The distance from zero to this interval is therefore a lower bound on the mean error attainable by expert reweighting. A positive confidence-adjusted distance should trigger abstention, probing, or structural investigation, not a claim that lower total mean-squared error repaired the blind spot. For vector-valued behavioral errors and consequence metric $Q \succeq 0$, the corresponding quantity is

$$\min_{w \in \Delta_E} \left(\sum_e w_e \mu_e \right)^\top Q \left(\sum_e w_e \mu_e \right).$$

This convex program is a diagnostic boundary between coalition selection and representation or mechanism repair; learning calibrated behavioral error vectors remains open.

5 Complementary memory

EOS separates working, episodic, semantic, procedural, and counterexample memory. This is computational in-

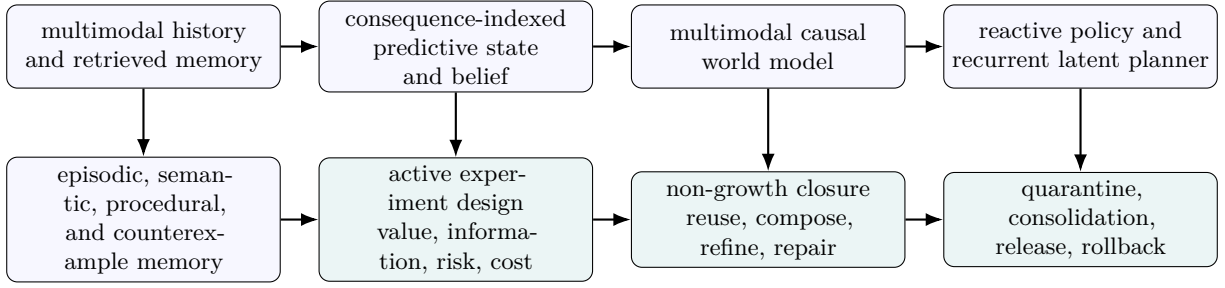


Figure 1: Prediction and authority are coupled but distinct. Ordinary behaviour uses the dense path. Surprise launches a bounded investigation; it cannot by itself authorize memory consolidation or structural growth.

spiration, not a literal brain model. Generative systems-consolidation models reconstruct episodes from complementary memory systems [18]; normative work suggests that unrestricted consolidation can harm generalization and that memories should consolidate when they improve future behaviour [19].

A consolidation proposal m is scored by

$$J(m) = \Delta L_{\text{future}}(m) + \eta \Delta V(m) - \lambda_C C(m) - \lambda_R R(m) - \lambda_D D(m), \quad (5)$$

where gains are measured on replay-separated or fresh data, C is resource cost, R is irreversible risk, and D is semantic distortion. A positive score proposes consolidation; scope and evidence still control release. Counterexample memory preserves rare contradictions specifically to prevent unsafe state merges and overgeneralization.

6 Transformers, JEPA, and neuromorphic systems

These are not mutually exclusive architectures:

- a **Transformer**, recurrent network, or state-space model is a sequence operator;
- a **joint-embedding predictive objective** specifies what a representation and predictor learn; and
- **digital, analogue, photonic, spiking, or spintronic hardware** specifies how selected operators are physically executed.

A hybrid system may train a joint-embedding predictive state, use Transformer attention for variable binding, recurrent depth for iterative computation, specialized mechanism modules for causal composition, analogue arrays for matrix operations, and stochastic devices for future sampling. The evidence and provenance path remains digital unless an approximate implementation earns the same audit guarantees.

Action-conditioned predictive representation learning also has close direct precedent. seq-JEPA separates invariant and equivariant representations by conditioning view sequences on relative transformations [7], while V-JEPA 2 post-trains an action-conditioned latent model for planning [1]. EOS must compare against these systems rather than relabel their ingredients as a new architecture.

Analogue in-memory systems have demonstrated kernel operations and robust hardware-aware training [3, 16]. Neuromorphic systems have advanced in scale but still face workload, training, software, and measurement challenges [9]. Spintronic devices offer nonvolatility, stochasticity, and physical dynamics, while connectivity, integration, and device sensitivity remain substantial [13]. These findings motivate heterogeneous co-design, not the claim that a brain-like substrate is sufficient for intelligence.

For implementation h of operator o , the hardware contract is the tuple

$$\Gamma(o, h) = (\text{accuracy, calibration, energy, latency, area, lifetime, conversion, communication, programmability}).$$

A substrate is preferred only if it Pareto-improves the declared deployment contract or satisfies a hard constraint. Device operations per joule are not a system result when conversion or memory movement dominates.

7 Training objectives

The proposed objective is multi-timescale:

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{predictive consequence}} + \lambda_b \mathcal{L}_{\text{belief calibration}} \\ & + \lambda_c \mathcal{L}_{\text{counterfactual}} + \lambda_r \mathcal{L}_{\text{retrieval}} \\ & + \lambda_a \mathcal{L}_{\text{algorithmic}} + \lambda_s \mathcal{L}_{\text{structural regret}} \\ & + \lambda_h \mathcal{L}_{\text{hardware cost}}. \end{aligned} \quad (6)$$

These terms should be treated as constraints or dual objectives rather than silently collapsed into one scalar. A gain in average prediction must not spend an undeclared semantic, calibration, or resource budget.

8 Falsification programme

The first three experiments precede broad scaling because an uncalibrated cause universe or aliased representation invalidates later structural claims. Hardware experiments use traces from scientifically valid software tasks. The final cross-domain test freezes algorithm, hyperparameters, and evidence budgets before moving to a new sensor and mechanism family.

Table 1: Ordered experiments. Each stage has a result that would weaken or reject the corresponding architectural hypothesis.

Experiment	decisive measurement	rejection condition
Cause-set coverage	fixed-sequence coverage, closure cost, false growth	set proposal offers no cost/power gain or excludes causes above its bound
Active disambiguation	state-split accuracy and value after probe cost	probes do not improve structural decisions or downstream value
Nonlinear closure	false/missed growth, modules, regret, compute	no advantage over dense, changepoint, MDL, or dynamic-expert baselines
Memory consolidation	rare recall, generalization, footprint	matched replay or retrieval dominates all three
Recurrent reasoning	length/composition generalization versus recurrence	matched Transformer/state-space model is better at equal compute
Scientific discovery	causal recovery and information per experiment	active hypotheses do not beat generic exploration
Precision frontier	semantic decisions under noise and quantization	approximate operators change governed decisions undetectably
Hybrid placement	end-to-end energy/latency at matched accuracy	conversion and control erase device-level gains
Continual world	integrated prediction, memory, control, growth	component gains disappear or interact destructively
Cross-domain freeze	transfer after freezing method and budgets	performance requires domain-specific redesign or test tuning

9 Current evidence and missing pieces

Controlled local studies already support narrow interfaces: fresh confirmation of learned consequence counterexamples; typed separation of reversible sensing from persistent mutation; lifetime spending in restricted binary settings; exact crossing-state quarantine; and a supplied causal closure whose omitted representation-repair family creates false novelty. These are component tests.

Recent controlled studies also expose two useful boundaries. A small RGB encoder learns coloured-marker identity and approximate camera coordinates, but a frozen visual state followed by learned geometric memory, GRU, state-space, and Transformer controls all remain at chance on a delayed object-axis query. Action-conditioned joint-embedding post-training keeps a non-collapsed state but improves zero-action latent prediction by only 3.81%, inverse-action decoding by 4.33%, and cross-background coordinate probing by 1.33%. These negative results motivate explicit invariant/equivariant factorization rather than larger memory. Separately, a scalar generated-council study detects all five common-bias controls with zero false triggers and a minimum conservative convex-hull distance of 0.7346, while mixed-sign cancelable bias remains below 0.00291. That certificate distinguishes denoising from shared-error repair but does not yet cover learned or vector-valued behavioral consequences.

The complete system remains unimplemented. There is no learned open-world cause ontology, pixel-based nonlinear causal closure, language-scale recurrent reasoner, broad-domain continual learner, autonomous scientific discovery, or end-to-end physical implementation. A serious architectural result requires the integrated and trans-

ferred experiments in Table 1, strong matched baselines, frozen evaluation, and independent reproduction.

10 Conclusion

The proposed architecture treats intelligence as prediction under controlled change. High-capacity world models provide the generative substrate; causal queries define what abstractions must preserve; active experiments separate hidden explanations; complementary memory supports rapid retention and slow generalization; recurrent latent operators provide adaptive computation; and a typed evidence governor controls persistent authority. Neuromorphic and analogue hardware enter as measured implementations of suitable operators, not as an independent theory of intelligence. The programme succeeds only if these interfaces improve learned nonlinear, integrated, and cross-domain systems over stronger and simpler alternatives.

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